

Recitation 9

Q1)

Suppose you are observing the bond market and notice two seemingly identical 5-year corporate bonds issued by the same company. Both bonds offer an annual coupon rate of 6% but are currently trading at different prices:

- Bond A is trading at \$980.
- Bond B is trading at \$1,050.

Both bonds have a face value of \$1,000 and will mature in 5 years.

Question: Analyzing Arbitrage Opportunity and Price–Return Relationship

- (a) Calculate the exact yield to maturity (YTM) for Bond A and Bond B. The YTM for each bond is the rate r that satisfies the equation:

$$P = \sum_{t=1}^n \frac{C}{(1+r)^t} + \frac{F}{(1+r)^n}$$

where:

- P is the bond's price.
 - C is the annual coupon payment ($\$1,000 \times 6\% = \60).
 - F is the bond's face value (\$1,000).
 - n is the number of years until maturity (5 years).
 - r is the YTM.
- (b) Based on the YTM calculations, identify the bond that should be purchased and the bond that should be sold to take advantage of an arbitrage opportunity.
- (c) Explain the relationship between the price and yield (return) of these bonds, illustrating the concept of inverse relationship.

Q2)

Consider an extension of the Solow model where the economy experiences exogenous technological progress at a constant rate g , with the production function given by:

$$Y_t = K_t^{0.5}(A_t L_t)^{0.5}$$

where A_t grows at rate $g = 0.02$ per period.

1. Effective Capital per Effective Worker Evolution.

Express the equation for the evolution of effective capital per effective worker in the presence of technological growth, where effective capital per effective worker is defined as $k_t = K_t/(A_t L)$ and effective output per effective worker is defined as $y_t = Y_t/(A_t L)$.

2. Steady-State Analysis.

Discuss how the steady-state level of effective capital per effective worker,

$$k^* = \frac{K}{A_t L},$$

is affected by technological progress. Derive the steady-state value of k^* .

3. Long-Run Growth Implications.

Explain the implications of continuous technological progress for long-term economic growth and per capita income growth.